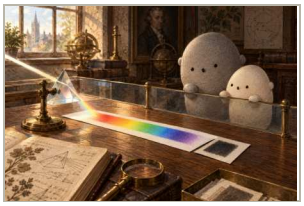
A little strip of light

The light we cannot see. · Curiosity day

Name _____ Date _____



A simplified reconstruction of Ritter's discovery, not a home experiment. The dark sample is a clue to unseen UV.

A SIMPLIFIED STRIP · POSITIONS ONLY

Light	UV-A	Violet	Blue	Green	Yellow	Orange	Red
Place	1	2	3	4	5	6	7

UV-A is unseen. The other six names are visible colors. Real colors blend; these boxes are not equal measured bands.

1. There are 7 places. One is unseen UV-A. How many places name visible colors?

$7 - 1 =$ _____ Show your thinking.

2. Start at place 1, UV-A. How many one-place hops reach place 7, red?

$7 - 1 =$ _____ Show your thinking.

3. Green is at place 4. Red is at place 7. How many places farther along is red?

$7 - 4 =$ _____ Show your thinking.

Wavelength is the space from one wave crest to the next. Imagine how stretched out a drawn wave looks. Along our strip toward red, wavelengths get longer. Which end would have more closely spaced crests?



WONDER DAILY

The light we cannot see.

Make a little book



1 Fold it into 8 boxes. Open it flat.



2 Fold short end to short end. Cut the thick line. Open it.



3 Fold the long way, pictures out. Push the ends in.



4 Fold into a book. Page 1 goes on the front!



Beyond violet

Their search begins long ago. In 1801, Johann Ritter spread sunlight into a rainbow.



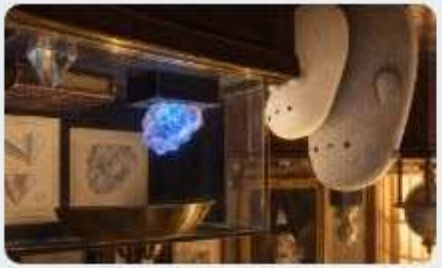
Two paper stars

A green star shines in a quiet museum case. Beside it, a cream star stays dull. Both sit under the same dark-looking lamp.



Keep the clue

The museum lamp switches off. The green fluorescence stops almost at once. Little sketches a lamp, a star, and light coming out.



A name for the glow

Another display tells of George Stokes. In 1852, he studied materials that took in light and gave out a different color.



Light in, light out

The green star has a special coating: a fluorescent phosphor. It absorbs UV-A and quickly emits green light.



A dark-looking light

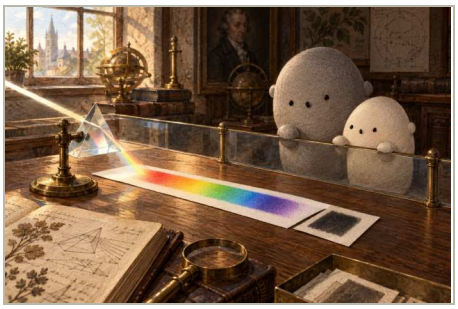
Back at the stars, Little studies the black light. It gives out mostly UV-A, the ultraviolet near visible violet. Our eyes cannot see UV-A.

Give the glow a path.

The light we cannot see. · Curiosity day

Name _____ Date _____

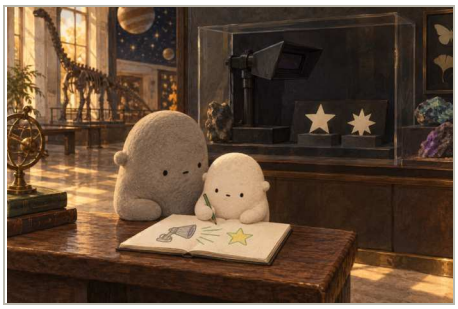
Choose one scene. Say what the clue helped Little discover. Draw, label, or write one idea.



Beyond violet



Light in, light out



Keep the clue

Draw a clue from your scene. Add an arrow to show where light goes or a label to name a change.

UV-A · phosphor · absorb · emit · fluorescence · visible · clue

The dark patch was a clue because … What was reaching it?

The coating takes in … What light goes in? What light comes out?

When the lamp goes off, … What happens to fluorescence?

Point to the detail your words explain. If you write a sentence, check its first capital and ending mark.

The light we cannot see.

01 Why does only one star glow?



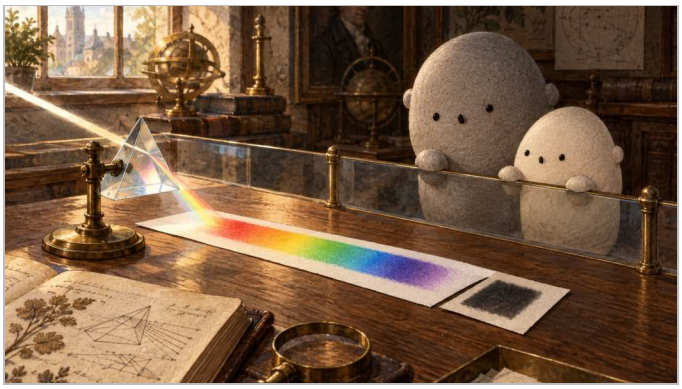
An imagined museum display. The stars have different coatings; their shared shape does not explain the glow.

Look closer. Find the two paper stars. What looks the same? What looks different? Keep a guess before opening the explanation.

A little explanation. The glowing star has a fluorescent coating. It takes in light from the black light and gives out green light we can see. The other star has no coating that makes a noticeable glow in this model. Shape alone does not decide which object glows.

Try this. Point to the part you would change to test your guess in a drawing: the lamp, the star's shape, or its material.

02 Can light reach a place our eyes cannot see?



A simplified reconstruction of Ritter's discovery, not a home experiment. The dark sample is a clue to unseen UV.

Look closer. Follow the rainbow to its violet end. The small dark patch sits just beyond it. What might a change in the paper tell us?

A little explanation. In 1801, Johann Ritter found that light-sensitive material changed beyond the violet end of a spread-out rainbow. Something unseen was reaching it. We call that light ultraviolet, or UV. UV-A is the part near visible violet that ordinary black lights use. UV-A itself is invisible to our eyes.

Try this. Trace the rainbow in the picture. Pause at violet, then point just beyond it. Imagine finding a clue without seeing what made it.

03 Is a black light really sending out darkness?



The violet tint is visible light at the lamp. UV-A itself cannot be seen; no real UV beam is pictured.

Look closer. Look at the lamp's dim violet face and the space beneath it. A picture cannot show invisible light directly. What could reveal its presence?

A little explanation. A black light sends out mostly UV-A light. A little visible violet can leak out too; that is the purple we see at the lamp. The invisible UV-A is not darkness. A suitable fluorescent material can help us notice what the lamp is doing.

Try this. Imagine putting the two drawn stars beneath this lamp. Tell which clue you would watch for. The screen model is enough.

04 How can a material make a new color of light?



The coating is enlarged for noticing. This is a material model, not a picture of atoms or an exact glow measurement.

Look closer. The lens enlarges the star's coating. Follow the idea from lamp to coating to green glow. Where is the visible light coming from?

A little explanation. A phosphor is a material that can take in energy and give out light. Here a fluorescent phosphor takes in UV-A and quickly gives out visible green light. Some fluorescent dyes do this too. The quick re-glow is called fluorescence. Different materials can glow in different colors, or show no glow under this lamp.

Try this. Tell a three-part path using these words: UV-A in, material, visible light out. Draw arrows if they help.